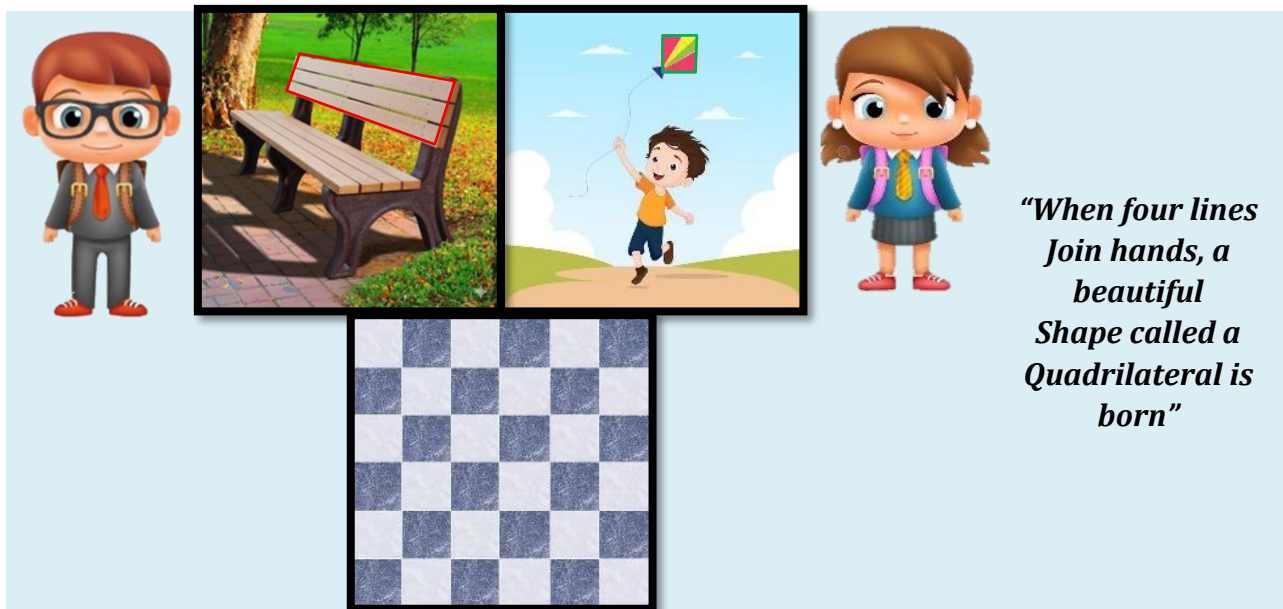


4

Quadrilaterals



*"When four lines
Join hands, a
beautiful
Shape called a
Quadrilateral is
born"*

Introduction

Four sided shapes around us

One sunny afternoon, Riya and Kabir were walking in the park.

Kabir: "Riya, our teacher said we need to learn about quadrilaterals, but I don't get it. What are they?"

Riya (smiling): "Let's go on a shape hunt! A quadrilateral is any shape with four sides."

They looked around the park.

Riya: "Look at that bench—it has a rectangle. It has four sides, and all the angles are 90° ."

Kabir: "What about the kite that boy is flying?"

Riya: "Great catch! That's a kite—it also has four sides, but only two pairs of equal sides next to each other."

Then they saw shape of tiles on the pathway.

Kabir: "These tiles look like a square."

Riya: "Yes! A square has four equal sides and four equal angles. It's a special type of rectangle."

Kabir: "So, all these are quadrilaterals?"

Riya: "Yes! As long as a shape has four sides, it's a quadrilateral. They just come in different types!"

Kabir: "Wow! Quadrilaterals are everywhere!"

They laughed and kept walking, spotting more shapes along the way.

Quadrilaterals are four-sided shapes—and they are all around us. Before understanding quadrilaterals, let discuss some basic terms related to shapes.

Curve

A plane figure formed by joining a number of points without lifting a pencil from the paper and without retracing any portion of the drawing other than single points is called a curve.

Examples:

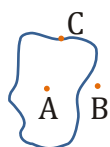


Curves can be of different types - open, closed, simple closed etc.

Region

Interior → Region inside the boundary of the closed curve.

Exterior → Region outside the boundary of the closed curve.

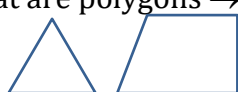


Point A lies in the interior region of the curve and point B lies in the exterior region of the curve. Here, point C lies on the curve.

Polygons

A polygon is a simple closed curve made of many line segments.

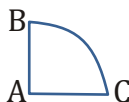
Examples of curves that are polygons →



Examples of curves that are not polygons are given below.



(not a closed curve)



(BC—not a line segment)



The polygon having no. of sides 20, 30, 100 and 1000 are called icosagon, triacontagon, hectagon and chiliagon respectively.

SPOT LIGHT



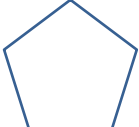
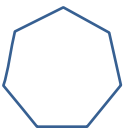
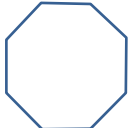
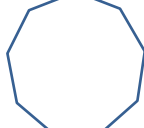
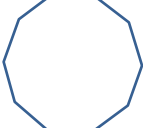
Do You

Remember ?

- ★ $0^\circ < \text{Acute angle} < 90^\circ$
- ★ $90^\circ < \text{Obtuse angle} < 180^\circ$
- ★ $180^\circ < \text{Reflex angle} < 360^\circ$

Classification

Polygons are classified by the number of their sides.

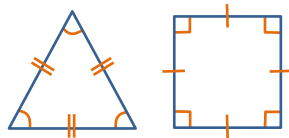
3 sides	4 sides	5 sides	6 sides
			
Triangle	Quadrilateral	Pentagon	Hexagon
7 sides	8 sides	9 sides	10 sidesn sides
			
Heptagon	Octagon	Nanagon	Decagon n-gon

Sides greater than 10 sides can be termed as n-sided polygon.

Regular and Irregular polygons

A Regular polygon is one that is both equilateral (all sides equal) and equiangular (all angles equal).

Equilateral triangle, square are equilateral and equiangular. So, they are Regular polygons.

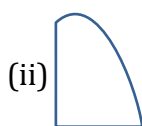
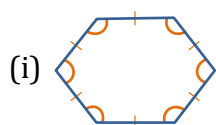


Rectangle is Equiangular but not equilateral.

So it is Irregular polygon.



Determine whether each figure is a polygon. If it is, classify the polygon and state whether it is regular. If it is not a polygon, explain why?



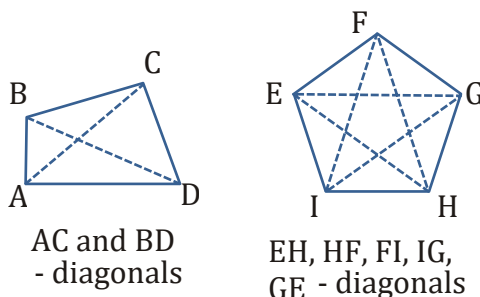
Explanation

(i) The figure has 6 equal sides and 6 equal angles. It is a regular hexagon.

(ii) The figure is not a polygon since it has a curved side.

Diagonals

A diagonal is a line segment connecting two non-consecutive vertices of a polygon.



So, in this chapter, we will explore various fascinating types of four-sided shapes and work on problems related to them. These shapes are generally called quadrilaterals. The term 'quadrilateral' comes from Latin, where 'Quadri' means four and 'latus' means sides.



Do You

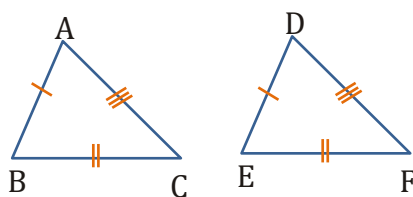
Remember?

★ Congruent figures

If two figures have the same shape and size, they are said to be congruent. For congruence, we use the symbol $\cong$. Two plane figures are congruent if each when superimposed on the other, covers it exactly.

★ Congruence of triangles

Two triangles are congruent if pairs of corresponding sides and corresponding angles are equal.

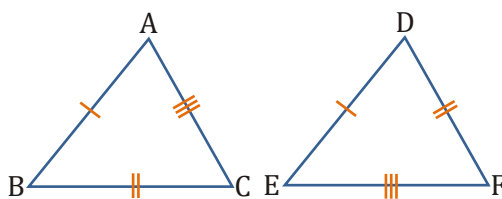


Thus, $\triangle ABC \cong \triangle DEF$, if $AB = DE$, $BC = EF$ and $CA = FD$, $\angle A = \angle D$, $\angle B = \angle E$, and $\angle C = \angle F$.

For the triangles to be congruent, the lengths of the corresponding sides and the measures of the corresponding angles of each of the triangles must be the same.

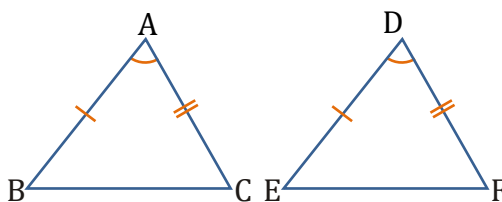
- ★ To ensure the congruence of two triangles, it is sufficient to verify that only three pairs (taken in particular order) of corresponding parts are equal.
- ★ There are five congruence criteria to determine the triangles are congruent:

Case 1: S.S.S. Congruence criterion (Side-Side-Side)



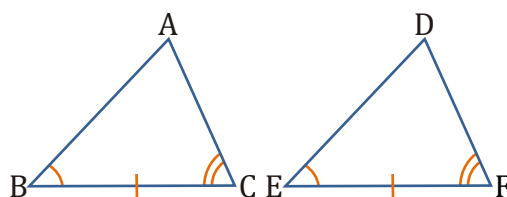
Two triangles are congruent if the three sides of one triangle are respectively equal to the three sides of the other triangle.

Case 2: S.A.S. Congruence criterion (Side-Angle-Side)



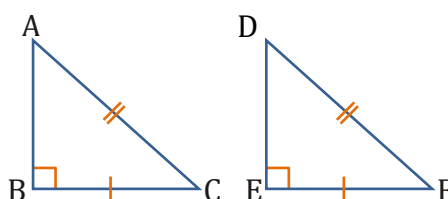
Two triangles are congruent if the two sides and the included angle of one triangle are respectively equal to the two sides and the included angle of the other triangle.

Case 3: A.S.A. congruence criterion (Angle-Side-Angle)



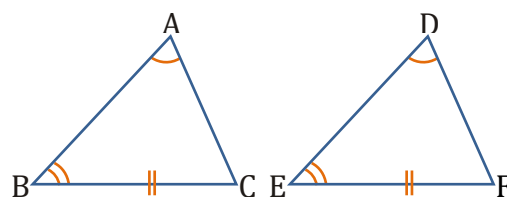
Two triangles are congruent if the two angles and the included side of one triangle are respectively equal to the two angles and the included side of the other triangle.

Case 4: R.H.S. congruence criterion (Right-Hypotenuse-Side)



Two right triangles are congruent if the hypotenuse and one side of the first triangle are respectively equal to the hypotenuse and one side of the second triangle.

Case 5: A.A.S. congruence criterion (Angle-Angle-Side)

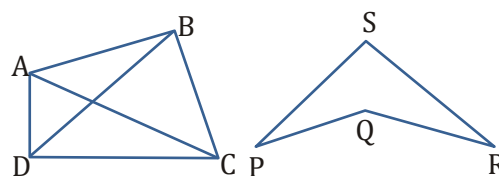


If two angles and a side of one triangle are equal to two corresponding angles and one side of the other triangle, then the triangles are congruent.

Quadrilaterals

A polygon with four sides and vertices is called a quadrilateral. In the figure given below, ABCD and PQRS are quadrilateral. Before learning different types of quadrilaterals, we must know some important terms related to quadrilaterals.

Vertex: A, B, C and D are vertices of the quadrilateral ABCD.



Sides: AB, BC, CD and DA are the sides of the quadrilateral ABCD.

Angles: $\angle A$, $\angle B$, $\angle C$ and $\angle D$ are the angles of the quadrilateral ABCD.

Adjacent sides: AB & BC (Common vertex B), BC & CD (Common vertex C), CD & DA (Common vertex D) and DA & AB (common vertex A) are the four pairs of adjacent sides in the quadrilateral ABCD.

Opposite sides: AB & DC and BC & AD are the two pairs of opposite sides of the quadrilateral ABCD.

Adjacent angles: $\angle A$ & $\angle B$ (common side AB), $\angle B$ & $\angle C$ (common side BC), $\angle C$ & $\angle D$ (common side CD) and $\angle D$ & $\angle A$ (common side AD) are the four pairs of adjacent angles in the quadrilateral ABCD.

Opposite angles: $\angle A$ & $\angle C$ and $\angle B$ & $\angle D$ are the two pairs of opposite angles in the quadrilateral ABCD.



Quick Tips

- ★ Any four-sided closed shape is a quadrilateral, but the sides have to be straight and it has to be 2-dimensional (2-D).

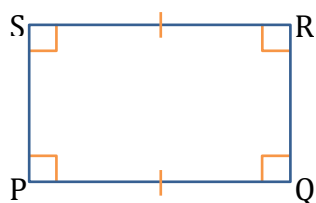
Types of Quadrilaterals

Rectangles

Properties of Rectangle

A rectangle is a quadrilateral in which-

- (i) The angles are all right angles (90°)
- (ii) The opposite sides are of equal length.

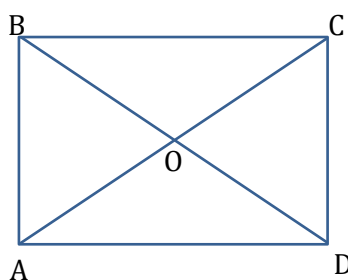


$$\angle P = \angle Q = \angle R = \angle S = 90^\circ$$

$$PQ = RS, SP = RQ$$

- (iii) The opposite sides are parallel to each other.
- (iv) Diagonals of a rectangle are equal.

In rectangle ABCD, diagonals $AC = BD$.



Proof: In $\triangle ADC$ and $\triangle DAB$

$$AB = CD$$

$$\angle BAD = \angle CDA = 90^\circ$$

$$AD = AD$$

(Common side)

$$\text{So, } \triangle ADC \cong \triangle DAB$$

(by SAS congruency)

$$AC = BD$$

(by cpct)

Hence, the Diagonals of a rectangle are equal.

- (v) Diagonals of a rectangle bisect each other.



The diagonals of a rectangle need not be perpendicular to each other.

SPOT LIGHT



**Quick
Tips**

- ★ Angle sum property of a triangle: The sum of the interior angles of triangle is 180° .
- ★ CPCT : Corresponding parts of congruent triangles.
- ★ Diagonals of a rectangle bisect each other bisecting means dividing a quantity into two equal parts.

In rectangle ABCD let point O be the intersection of two diagonals. Then, $AO = CO$ and $BO = DO$.

Proof : In $\triangle AOB$ and $\triangle COD$

$$\angle AOB = \angle COD \quad (\text{Vertically opposite angles})$$

$$\angle AOB = 90^\circ - \angle CBO$$

$$\text{and } \angle CDO = 180^\circ - (90^\circ + \angle CBO) = 90^\circ - \angle CBO$$

$$\therefore \angle ABO = \angle CDO \quad (90^\circ - \angle CBO)$$

$$AB = CD$$

So, by AAS congruency, $\angle AOB \cong \angle COD$

Hence, $AO = CO$

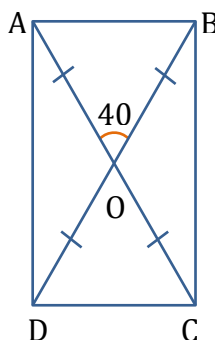
$$BO = DO \quad (\text{By CPCT})$$



**Building
Concepts**

2

Find the remaining angles of the given quadrilateral.



Explanation

Given, $\angle AOB = 40^\circ$

$$\therefore \angle DOC = 40^\circ \quad (\text{Vertically opposite angles})$$

$$\angle AOB + \angle BOC = 180^\circ \quad (\text{Linear pair})$$

$$\therefore \angle BOC = 180^\circ - 40^\circ = 140^\circ$$

$$\angle AOD = 140^\circ \quad (\text{Vertically opposite angles})$$

Now in $\triangle AOB$, $AO = OB$

$$\Rightarrow \angle BAO = \angle ABO = a$$

$$a + a + 40^\circ = 180^\circ \quad (\text{angle sum property})$$

$$2a + 40^\circ = 180^\circ$$

$$a = \frac{180^\circ - 40^\circ}{2} = 70^\circ$$

Similarly, In $\triangle DOC$, $DO = OC$

$$\Rightarrow \angle ODC = \angle OCD = 70^\circ$$

Now, in $\triangle BOC$, $BO = OC$

$$\Rightarrow \angle OBC = \angle OCB = b$$

$$b + b + 140^\circ = 180^\circ$$

$$b = \frac{40}{2} = 20^\circ$$

Similarly, in $\triangle AOD$, $AO = OD$

$$\Rightarrow \angle OAD = \angle ODA = 20^\circ$$

Clearly in the figure, we can see that

$$\angle AOB \cong \angle COD \quad \text{by SAS criterion}$$

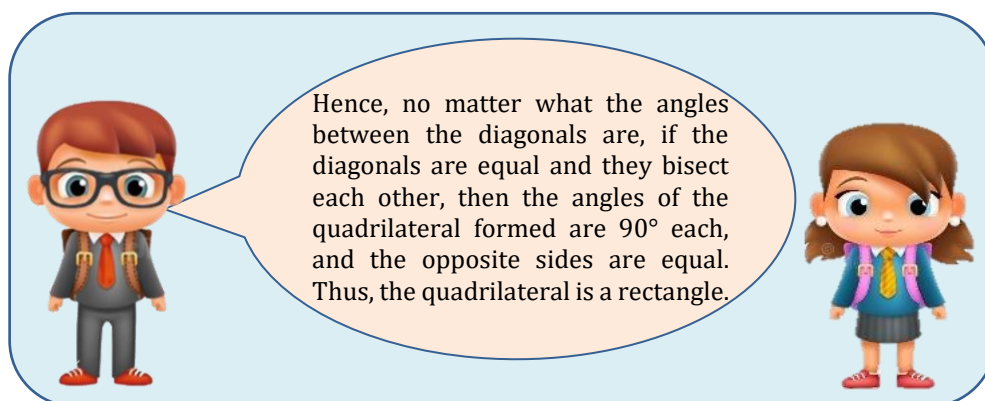
$$\text{So, } AB = CD \quad (\text{By CCPT}) \quad \dots (1)$$

$$\text{and } \triangle AOD \cong \triangle COB \quad \text{By SAS criterion}$$

$$\text{So, } AD = CB \quad (\text{By CPCT}) \quad \dots (2)$$

From equation (1) and (2)

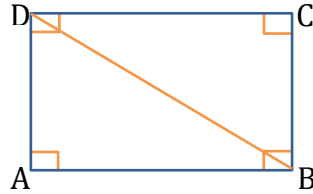
We can say that quadrilateral ABCD is a rectangle.





What is the shape of a quadrilateral with all the angles equal to 90° ?

Explanation



Let ABCD be a quadrilateral in which $\angle A = \angle B = \angle C = \angle D = 90^\circ$

Now, join BD

In $\triangle ADB$ and $\triangle CBD$

$\angle DAB = \angle BCD$ ($\therefore 90^\circ$)

$DB = BD$ (Common)

$\angle ADB = \angle CBD$ (Equal to $90^\circ - \angle CDB$)

$\therefore \triangle ADB \cong \triangle CBD$ (By AAS criterion)

Hence, $AD = CB$... (1)

$AB = CD$ (BY CPCT) ... (2)

So, From Eq. (1) and (2), given quadrilateral ABCD is a rectangle.



Thus, we have established that if all the angles of a quadrilateral are right angles, then the opposite sides have equal lengths. Therefore, the quadrilateral is a rectangle.



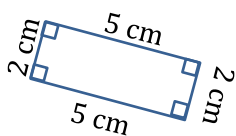
Square



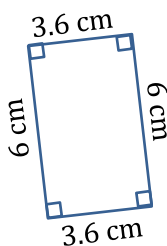
"Hey, you see all these shapes given below? They're all rectangles — even this one, figure- (iv). It's definitely a rectangle because all its angles are 90° . But you know what's cool? This one's special... all its sides are of the same length. So yeah, it's a rectangle, but it's actually a square too!"



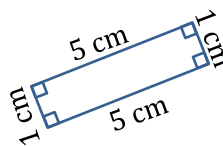
A square is a quadrilateral in which all the angles are equal to 90° , and all the sides are of equal length.



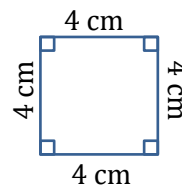
(i)



(ii)



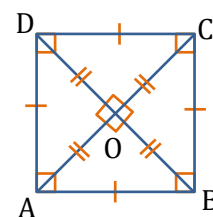
(iii)



(iv)

Properties of square

- (i) All the sides of a square are equal. Here, $AB = BC = CD = DA$
- (ii) The opposite sides of a square are parallel to each other. Here, $AB \parallel CD$ and $AD \parallel CB$
- (iii) The angles of a square are all 90° . Here, $\angle A = \angle B = \angle C = \angle D = 90^\circ$.
Here, $AC = BD$, $OA = OB = OC = OD$, $\angle AOB = \angle BOC = \angle DOA = 90^\circ$.
- (iv) The diagonals of a square are of equal length, and they bisect each other at 90° .
- (v) The diagonals of a square divide the angles of the square into equal halves. This can also be expressed as-The diagonals of a square bisect the angles of the square.



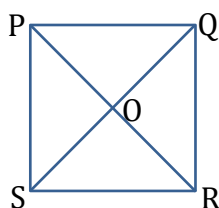
Here, $\angle ADB = \angle CDB$



Prove that:

- (i) The angles formed by the diagonals of square intersect at 90° .
- (ii) The diagonals of a square bisect the angles of the square.

Explanation



square PQRS,

$$PQ = QR = RS = SP$$

Diagonals of square are equal and bisect each other.

So, $PO = OR$... (i)

Now, In $\triangle POQ$ and $\triangle ROQ$

$PQ = OR$ {from (i)}

$QO = QO$ {common side}

$\angle POQ \cong \triangle ROQ$ {by sss congruency}

$\angle POQ = \angle QOR$ {by cpct}

So, $\angle POQ + \angle QOR = 180^\circ$

$\angle POQ = 90^\circ = \angle QOR$

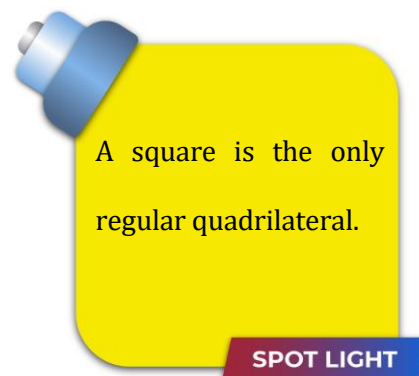
(i) So, angles formed by the diagonals of square are 90° .

(ii) Also by cpct, $\angle PQO = \angle RQO$

So, diagonal SQ intersect $\angle Q$

Similarly we can prove diagonal SQ intersect $\angle S$. And diagonal PR bisect $\angle P$ and $\angle R$.

Hence, diagonals of a square bisect the angles of square.



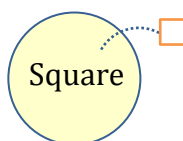
Be Alert !

- ★ Thus, every square is also a rectangle, but clearly every rectangle is not a square.
- ★ This relation can be pictorially represented using a Venn diagram.

Venn diagram

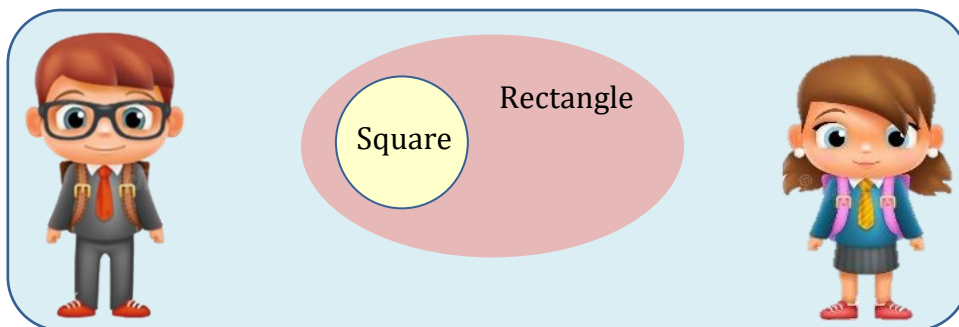
Kabir: Hey, can you tell me what a Venn diagram is?

Riya: It's a diagram that uses closed curves (either oval or circles) to show how different sets of things are related or overlap.



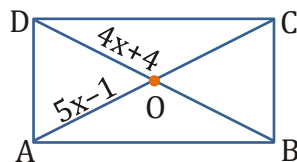
Here, each point in the region represents a square, thereby covering all the possible squares.

Venn representation of two sets of squares and rectangles is as follows:





In the given rectangle ABCD, the diagonals AC and BD meet at O.
Find x, if $OA = 5x - 1$ and $OD = 4x + 4$.



Solution

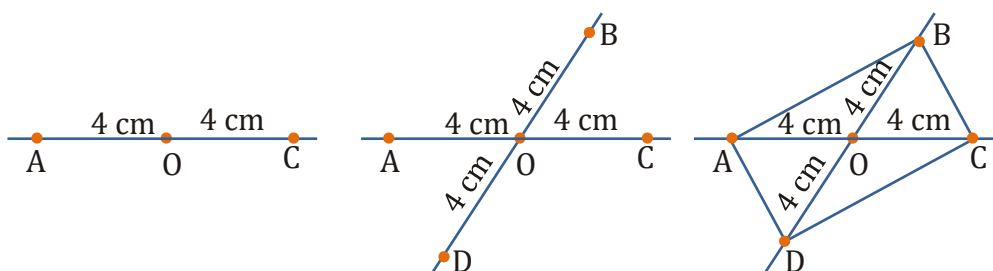
OA is half the diagonal AC and OD is half the diagonal BD.

Since, the diagonals of a rectangle are equal, their halves must also be equal.

Hence, $OA = OD$

$$\Rightarrow 5x - 1 = 4x + 4$$

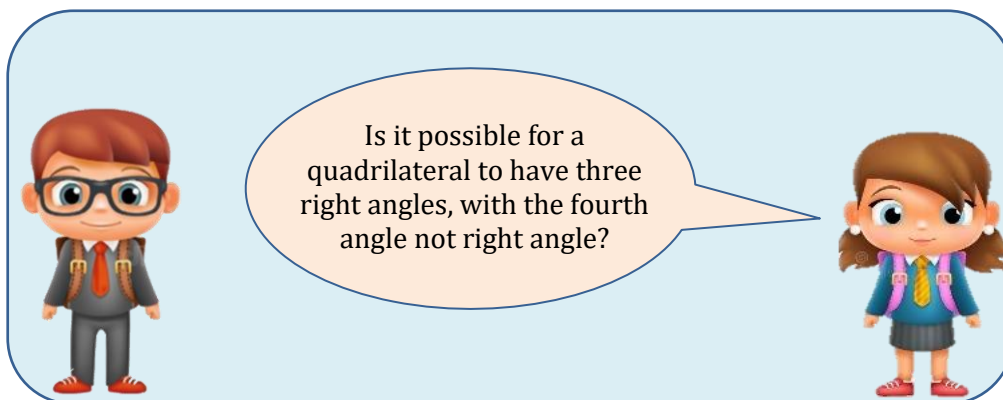
$$\Rightarrow 5x - 4x = 4 + 1 \Rightarrow x = 5$$



We know how the wooden strips have to be put together to form the vertices of a rectangle! They should be equal and connected at their midpoints.

This method is actually used in practice to make rectangles. Carpenters in Europe use this method to get a rectangular frame. It is also known that farmers in Mozambique, a country in Africa, use this method while constructing houses to get the base of the house in a rectangular shape.

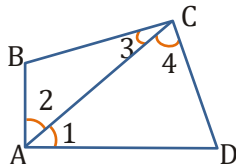
Angles in a Quadrilateral



Angle Sum Property of Quadrilaterals

The sum of all angles in any quadrilateral is 360° .

Proof : Let ABCD be a quadrilateral. Draw one of its diagonals, AC



Clearly, $\angle A = \angle 1 + \angle 2$ and $\angle C = \angle 3 + \angle 4$

In $\triangle ABC$, $\angle 2 + \angle 3 + \angle B = 180^\circ$ (Angle sum property of a triangle) ... (i)

In $\triangle ADC$, $\angle 1 + \angle 4 + \angle D = 180^\circ$ (Angle sum property of a triangle) ... (ii)

Adding (i) and (ii), we get

$$\angle 2 + \angle 3 + \angle B + \angle 1 + \angle 4 + \angle D = 180^\circ + 180^\circ$$

$$\Rightarrow \angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle B + \angle D = 360^\circ$$

$$\Rightarrow \angle A + \angle C + \angle B + \angle D = 360^\circ$$

This explains why a quadrilateral can't have three right angles, with the fourth angle not right angle.



The three angles of a quadrilateral are 75° , 105° and 85° . Find its fourth angle.

Solution

Let the fourth angle of the quadrilateral be x . According to the angle sum property of a quadrilateral,

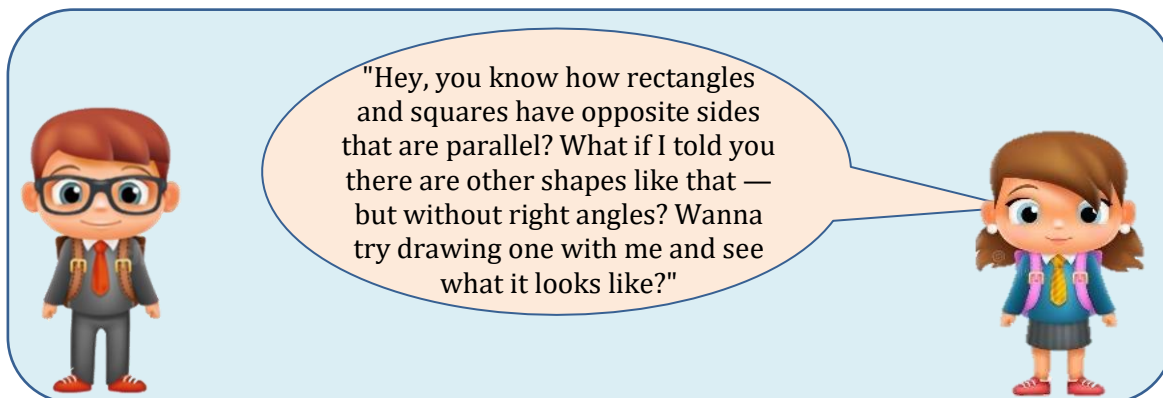
$$75^\circ + 105^\circ + 85^\circ + x = 360^\circ$$

$$\Rightarrow 265^\circ + x = 360^\circ$$

$$\Rightarrow x = 360^\circ - 265^\circ$$

$$\Rightarrow x = 95^\circ$$

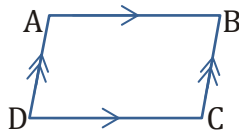
More Quadrilaterals with Parallel Opposite Sides



Parallelogram

A parallelogram is a quadrilateral in which both the pairs of opposite sides are parallel.

Quadrilateral ABCD is a parallelogram where $AB \parallel CD$ and $BC \parallel AD$.

**Properties of Parallelogram**

- (i) The opposite sides of a parallelogram are equal in length.

Proof: Consider a parallelogram ABCD.

Draw any diagonal, say AC.

Looking at the angles, $\angle 1 = \angle 4$.

(as $AB \parallel CD$ and AC is a transversal $\angle 1$ and $\angle 4$ are alternate interior angles.)

Similarly, $\angle 2 = \angle 3$

In triangles ADC and ABC, $\angle 1 = \angle 4$

$AC = AC$ (common side), $\angle 2 = \angle 3$

So, $\triangle ADC \cong \triangle CBA$

(ASA congruence rule)

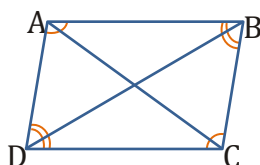
By CPCT,

$\Rightarrow CD = AB$ and $AD = BC$

Hence proved.

- (ii) The opposite angles in a parallelogram are equal.

Proof: We have proved above that $\triangle ADC \cong \triangle CBA$



$\Rightarrow \angle B = \angle D$

Similarly, we can draw the other diagonal BD and prove $\triangle DAB \cong \triangle BCD$

$\Rightarrow \angle A = \angle C$

Hence proved.

- (iii) The adjacent angles in a parallelogram are supplementary.

Proof: $AB \parallel CD$ and AD is a transversal, $\angle A$ and $\angle D$ are interior angles on the same side of the transversal.

$\Rightarrow \angle A + \angle D = 180^\circ$

Similarly, $\angle B + \angle C = 180^\circ$

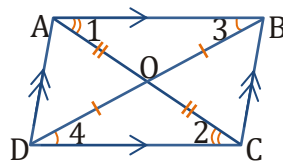
$\angle C + \angle D = 180^\circ$

$\angle D + \angle A = 180^\circ$



(iv) The diagonals of a parallelogram bisect each other.

Proof: Let us draw the two diagonals AC and BD (meeting at O) of the parallelogram ABCD. In triangles AOB and COD,



[Alternate interior angle with $\angle AB \parallel CD$ and $\angle AC$ -transversal.]

$\angle 1 = \angle 2$ (alternate interior angle)

$\angle 3 = \angle 4$ (alternate interior angle)

$CD = AB$ (opposite sides of parallelogram are equal)

So, $\triangle AOB \cong \triangle COD$ (ASA congruency)

$\therefore$ By CPCT,

$\Rightarrow AO = CO$ and $OB = OD$

Hence proved.



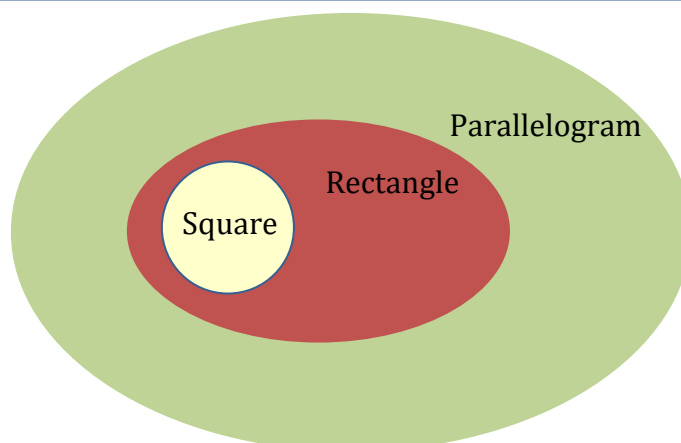
Quick Tips

★ In parallelogram ABCD, O is the midpoint of both the diagonals of given parallelogram.



Yes, we can represent by forming a Venn diagram as follows:

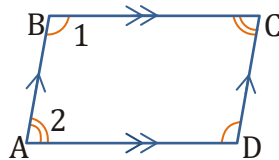
"Hey, can I ask you something? If a rectangle has opposite sides that are parallel, doesn't that mean it's a kind of parallelogram too? I mean, all its angles are 90° , so does that make it like a special type of parallelogram?"





Draw a parallelogram. Label the congruent angles.

Explanation



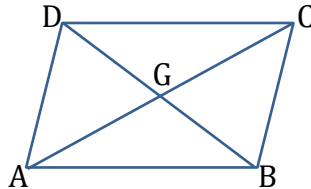
ABCD is a parallelogram with $AB \parallel CD$ and $BC \parallel AD$. We know, the opposite angles of a parallelogram are equal.

$$\Rightarrow \angle A = \angle C \text{ and } \angle B = \angle D$$

The 2 set of congruent angles



Complete the statements about parallelogram ABCD. Justify your answer.



(i) $\angle DAB \cong ?$

(ii) $\angle ABD \cong ?$

(iii) $AB \parallel ?$

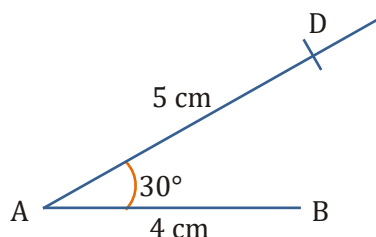
(iv) $BG = ?$



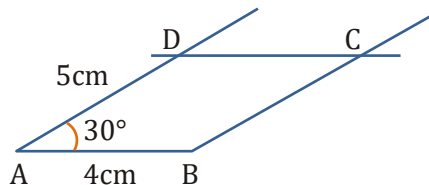
Draw a parallelogram with adjacent sides of lengths 4 cm and 5 cm, and an angle of 30° between them.

Explanation

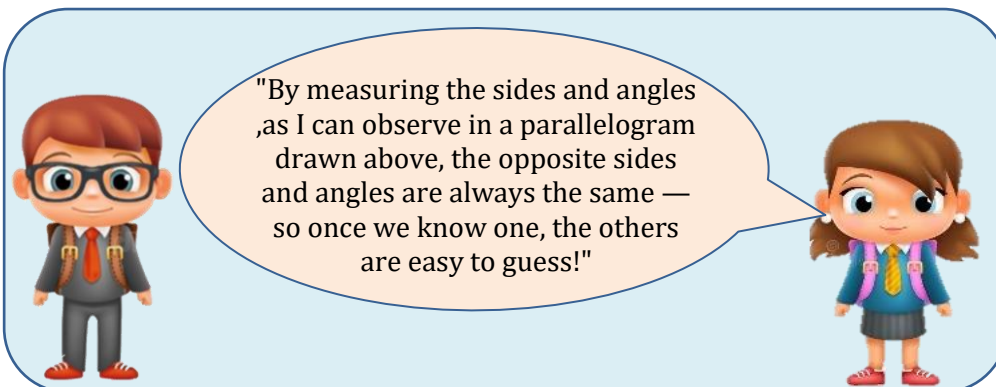
Step 1: Draw line segments $AB = 4$ cm and $AD = 5$ cm with an angle of 30° between them.



Step 2: Draw a line parallel to AB through the point D, and a line parallel to AD through B. Mark the point at which these lines intersect as C.

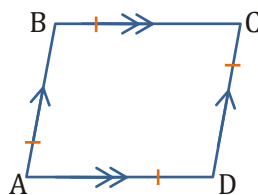


ABCD is the required parallelogram.



Quadrilaterals with Equal Side lengths

Rhombus



A rhombus is a parallelogram with all sides equal.

ABCD is a rhombus where $AB \parallel CD$ and $BC \parallel AD$.

Also, $AB = BC = CD = DA$.

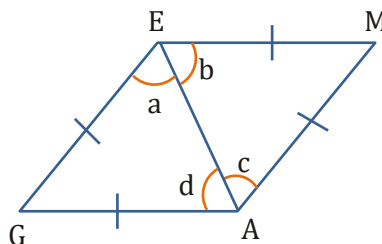
Rhombus has all the properties of a parallelogram. Besides these, a rhombus has some other properties also.

Properties of Rhombus

A quadrilateral in which all the sides have the same length is a rhombus.

- (i) All the sides of a rhombus are equal to each other.
- (ii) The opposite sides of a rhombus are parallel to each other.
- (iii) In a rhombus, the adjacent angles add up to 180° , and the opposite angles are equal.

Proof: Opposite angles are equal i.e. $\angle G = \angle M$, $\angle A = \angle E$



In $\triangle GAE$ and $\triangle MAE$

$$GE = GA$$

(Equal sides of a rhombus)

$$ME = MA$$

(Equal sides of a rhombus)

$$AE = AE$$

(common side)

$$\triangle GAE \cong \triangle MAE$$

(by SSS congruency)

$$\text{So, } \angle G = \angle M$$

(by cpct)

Similarly, $\angle GEA = \angle MEA$ or $a = b$

$$\angle GAE = \angle MAE \text{ or } d = c$$

But in $\triangle GAE$, $a = d$ and in $\triangle MAE$, $b = c$

(Angles opposite to equal sides are equal)

So, we can conclude $a = b = c = d$

$$\text{or } a + b = d + c$$

$$\angle E = \angle A$$

Hence, proved.

(iv) The diagonals of a rhombus bisect each other.

(v) The diagonals of a rhombus bisect its angles.

(vi) Diagonals of a rhombus intersect each other at an angle of 90° .

Proof: In $\triangle GEO$ and $\triangle MEO$

$$GE = EM$$

(Equal sides of a rhombus)

$$GO = MO$$

(Diagonals bisect each other)

$$EO = EO$$

(common side)

$$\triangle GEO \cong \triangle MEO$$

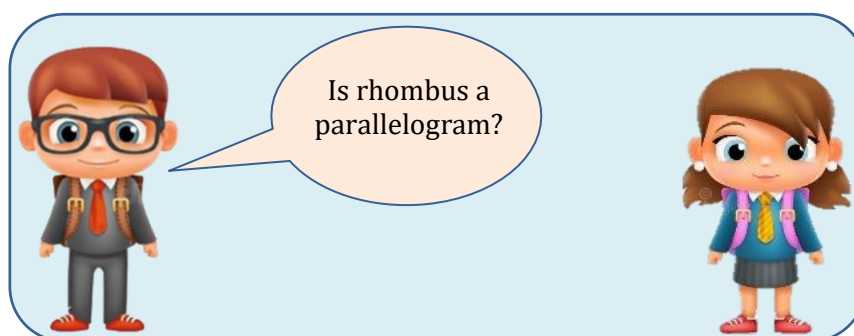
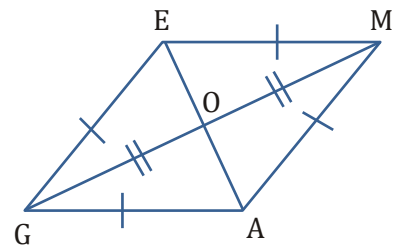
(by SSS congruency)

$$\text{So, } \angle GOE = \angle MOE$$

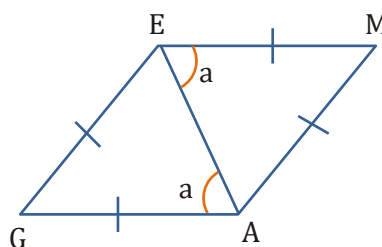
(by cpct)

But, $\angle GOE$ and $\angle MOE$ adds up to 180° ,

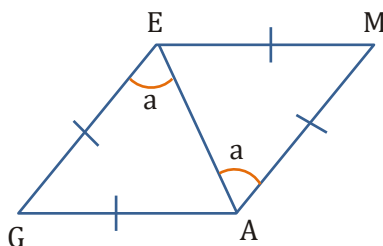
$$\text{So, } \angle GOE = \angle MOE = 90^\circ$$



Consider the lines EM and GA and its transversal AE. Since the alternate angles are equal, $EM \parallel GA$.

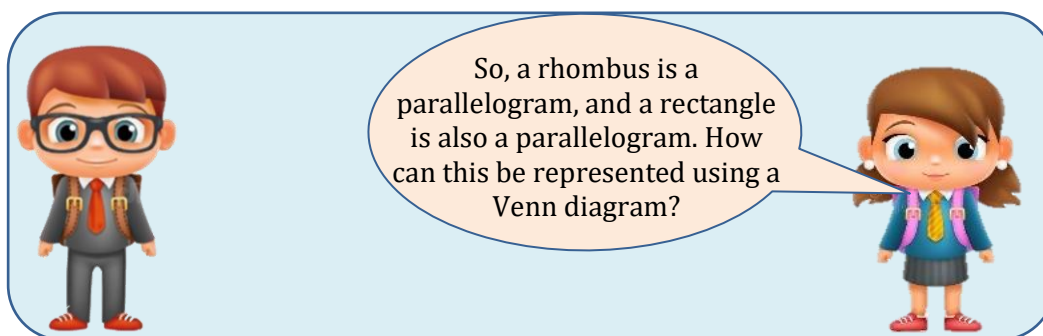


Similarly consider the lines GE and AM and its transversal AE. Since the alternate angles are equal, $GE \parallel AM$.



As opposite sides are parallel, GAME is also a parallelogram.

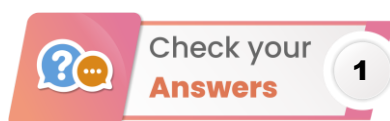
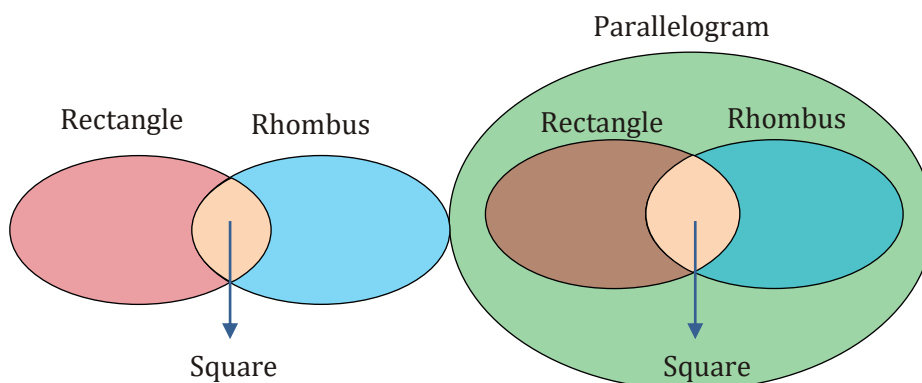
Thus, every rhombus is a parallelogram, and the properties of a parallelogram hold true for a rhombus as well.



Be Alert !

★ The diagonals of rhombus are not equal.

Venn Diagram will be as follows:



(i) $\angle DAB = \angle DCB$

(ii) $\angle ABD = \angle BDC$

(iii) $AB \parallel CD$

(iv) $BG = DG$

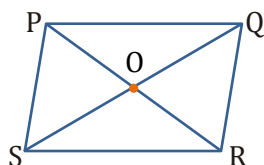


Which property is not a characteristic of Rhombus?

- (i) Opposite sides parallel (ii) 4 Right angles
(iii) 4 congruent sides (iv) Opposite sides congruent



In the given parallelogram PQRS, OS = 5 cm and PR is 7 cm more than QS. Find OP.



Solution

$OS = 5 \text{ cm} \Rightarrow OQ = 5 \text{ cm}$ (Diagonals of a parallelogram bisect each other.)

$\therefore QS = 10 \text{ cm}$

Given that $PR = QS + 7 = 10 + 7 = 17 \text{ cm}$

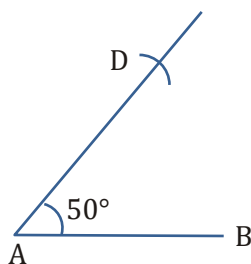
$\Rightarrow OP = \frac{1}{2} \times PR = \frac{1}{2} \times 17 = 8.5 \text{ cm}.$



Draw a quadrilateral such that all sides are equal and one angle is 50° .

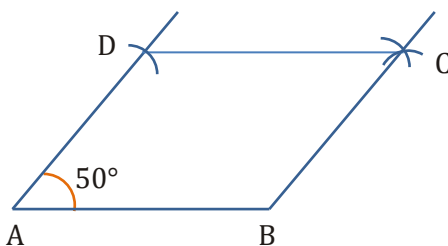
Explanation

Step-1 : Draw two equal sides AD and AB, that are forming 50° to each other.

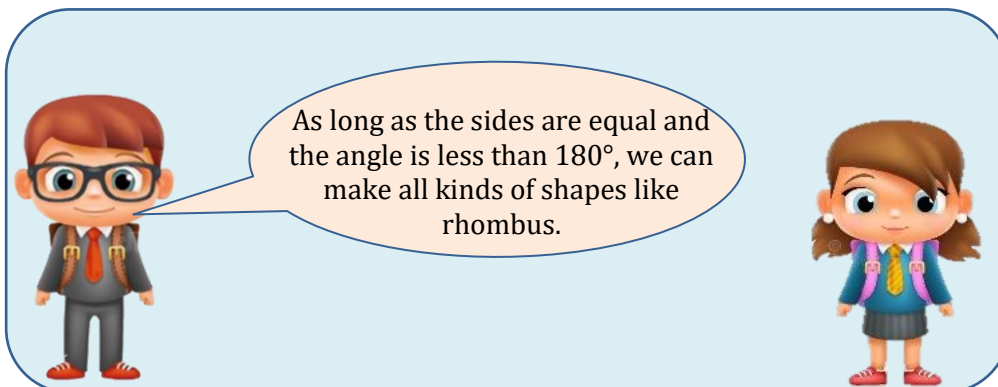


Step-2 : Mark a point C whose distance from B and D is equal to AB (or AD).

To do this, measure AB using a compass. Keeping this length as the radius, cut arcs from B and D.

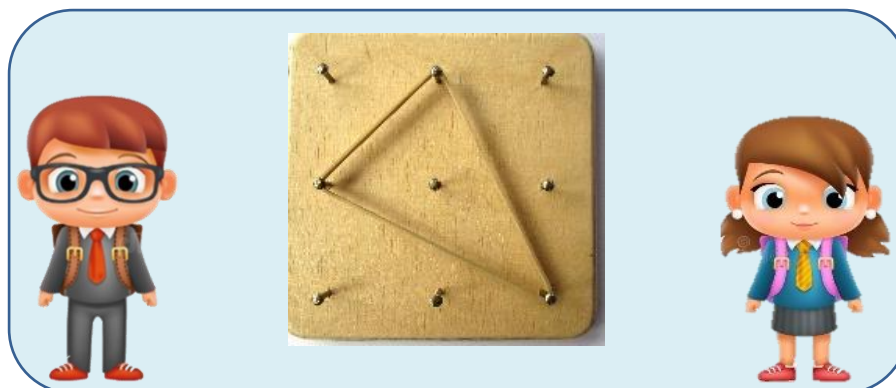


Step-3 : The quadrilateral formed is rhombus.



Playing with Quadrilaterals

Geoboard Activity



Kabir: Hey, I have a geoboard and some rubber bands. What can we make with them?

Riya: Great! Let's try an activity. First, place two rubber bands diagonally across the board so that they cross each other at the center and are of equal length. Make sure they are perpendicular to each other, like a big "+".

Kabir: Done! They form diagonals. Now what?

Riya: Now, join the ends of the rubber bands to form a closed shape. What you get is a ____ (P) ____ (square/rectangle)

Kabir: A ____ (Q) ____? Why? (square/rectangle)

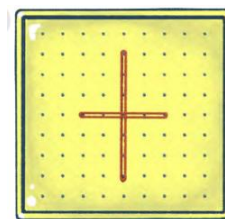
Riya: Yes! Because in this shape:

- All four sides are of ____ (R) ____ length. (equal/unequal)
- The diagonals bisect each other at ____ (S) ____ angles. (right/acute)
- The diagonals are of equal length and cross at the center.

This is a property of a ____ (T) ____, (square/rectangle), so it's a ____ (U) ____.

Answer:

P = square, Q = Square, R = equal, S = Right, T = Square, U = Square



Kabir: Okay, now what if I extend one of the diagonals by 2 cm on both sides?

Riya: Good! After extending, connect the new ends to the original shape to form a new quadrilateral.

Kabir: Done! What is this new shape?

Riya: Now you have a __ (P) __. (square/rhombus)

Kabir: Why is it a __ (Q) __? (square/rhombus)

Riya: Because:

- All four sides are of __ (R) __ length. (equal/unequal)
- The diagonals bisect each other at __ (S) __ angles. (right/acute)
- The diagonals are of unequal length and cross at the center.

This is a property of __ (T) __. (square/rhombus)

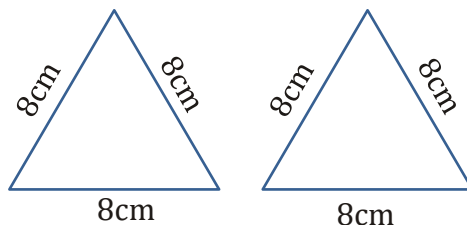
So by extending one diagonal, you've changed the __ (U) __ (square/rhombus) into a __ (V) __. (square/rhombus)

Answer:

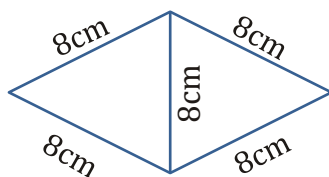
P = rhombus, Q = rhombus, R = right, S = rhombus, T = square, U = square, V = rhombus

Joining Triangles

Kabir: Hey, I have two cardboard cutouts of an equilateral triangle with each side 8 cm. Can you join them to make a quadrilateral?



Riya: Yeah! If you join them along one side, you get a quadrilateral.



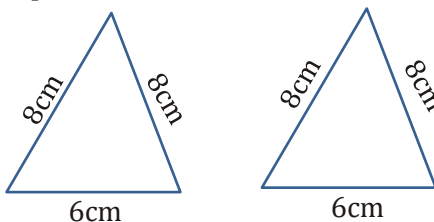
Since both triangles are equilateral, when joined,

- the shape will have sides __ (P) __. (equal/unequal)
- two pair of angles __ (Q) __. (equal/unequal) each will be __ (R) __. ($60^\circ/45^\circ$) and __ (S) __. ($120^\circ/35^\circ$)
- making it a kind of __ (T) __. (rhombus/square)

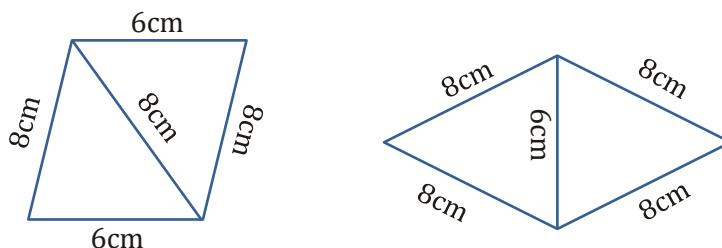
Answer:

P = equal, Q = equal, R = 60° , S = 120° , T = rhombus

Kabir: Cool! Now, I have two isosceles triangles with sides 8 cm, 8 cm, and 6 cm. In how many ways can we join them to form a quadrilateral?



Riya: Let's see! You can join them by matching the equal sides or the unequal sides. For example, joining them along the 8 cm sides or along the 6 cm sides. Each way you join them creates a different quadrilateral.



Kabir: What kind of quadrilaterals do they form?

Riya: If you join them along the equal sides (i.e. 6 cm), the quadrilateral will have two pairs of opposite sides equal, making it a ___(P)___ (rectangle/||gm)

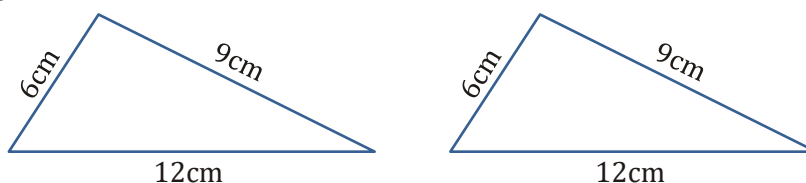
But if you join them along the unequal sides (i.e. 8 cm), the shape will look different, maybe a ___(Q)___ (parallelogram/rhombus) because the sides all be equal.

So, the shape depends on how the triangles are connected.

Answer:

P = ||gm, Q = rhombus

Kabir: Now, what if I take two scalene triangles with sides 6 cm, 9 cm, and 12 cm? Can they form a quadrilateral too?



Riya: Yes, they can! But since the sides are all different, the quadrilaterals formed will be ___(P)___ (rectangle/square/parallelogram/none).

We can join them in a few ways, like matching the longest sides or the shortest sides, but the angles and sides will all ___(Q)___ (equal/unequal)

Answer:

P = none, Q = unequal.

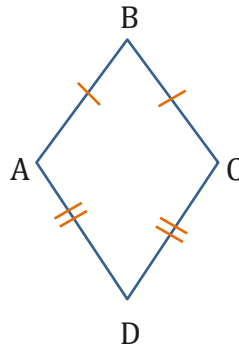
Kabir: Got it! So, by joining different triangles, we can create different types of quadrilaterals depending on how the sides match up and what lengths they have.

Riya: Exactly! It's like putting puzzle pieces together — the way you join them decides the shape you get.

Kite and Trapezium

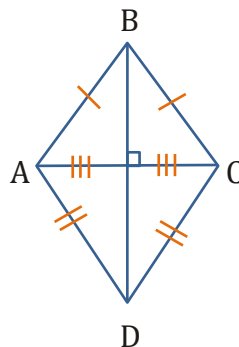
Kite

Kite is a special type of Quadrilateral in which one pair of adjacent sides is equal to each other and the other pair of adjacent sides are equal.



In the kite ABCD, $AB = BC$ and $CD = AD$

Properties of Kite

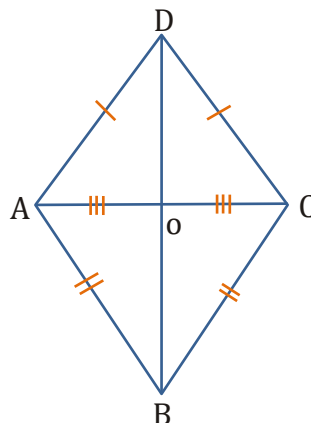


- (i) The diagonals are perpendicular to one another.
- (ii) One of the diagonals bisects the other. In the figure, BD bisects AC.
- (iii) One pair of opposite angles (the ones that are between the sides of unequal length) are equal in size. In the figure, $\angle A = \angle C$ but $\angle B \neq \angle D$.



In the kite, show that the diagonal BD

- (i) bisects $\angle ABC$ and $\angle ADC$,
- (ii) bisects the diagonal AC, that is, $AO = OC$, and is perpendicular to it.



Solution

In $\triangle ADC$

$$\Rightarrow AD = DC$$

$$\Rightarrow \angle ADO = \angle DCO \quad \dots(i)$$

(angles opposite to equal sides are equal)

Now, In $\triangle ADB$ and $\triangle CDB$

$$AD = DC$$

(by definition of kite)

$$AB = BC$$

$$BD = BD$$

(common side)

$$\triangle ADB \cong \triangle CDB$$

(by sss congruency)

$$\angle ADB = \angle CDB \quad \dots(ii)$$

(by cpct)

So, In $\angle AOD$ and $\angle COD$

$$AD = DC$$

$$\angle DAO = \angle DCO$$

[from (i)]

$$\Rightarrow \angle AOD \cong \angle COD$$

[by ASA congruency]

$$(i) \text{ By cpct, } \angle ADO = \angle CDO$$

$$\Rightarrow \text{Diagonal BD bisects } \angle ADC$$

Similarly, we can prove, $\angle ABO = \angle CBO$.

$$\Rightarrow \text{Diagonal BD bisects } \angle ABC.$$

$$(ii) \text{ By cpct, } AO = OC$$

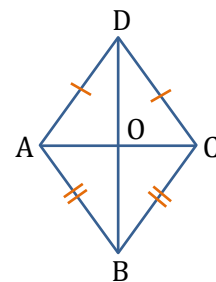
$$\text{and } \angle AOD = \angle COD$$

$$\Rightarrow \angle AOD + \angle COD = 180^\circ$$

$$\angle AOD = 90^\circ$$

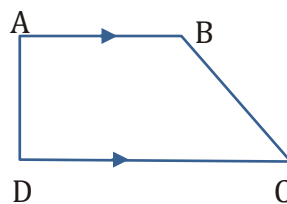
$$\angle AOD = 90^\circ = \angle COD$$

So, diagonal BD bisects diagonal AC and is perpendicular to it.



Trapezium

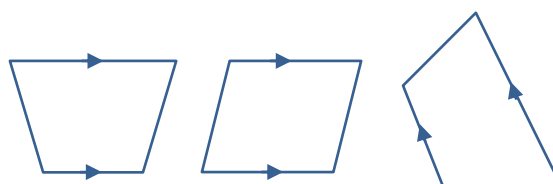
A quadrilateral in which a pair of opposite sides are parallel is called a trapezium.



$$AB \parallel CD$$

$$\Rightarrow ABCD \text{ is a trapezium.}$$

Other examples:



(Parallel sides denoted by Arrows)

Properties of Trapezium

In a trapezium, the interior angles on the same side of each of the non-parallel sides are supplementary.

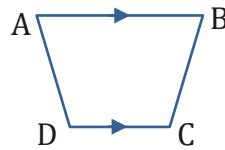
$$\therefore \angle A + \angle D = 180^\circ \quad (\text{as } AB \parallel CD \text{ and } AD \text{ is a transversal}).$$

$\angle A$ and $\angle D$ are interior angles on the same side of the transversal)

Similarly, $\angle B + \angle C = 180^\circ$

Isosceles trapezium

If the non-parallel sides of a trapezium are equal, then it is called an isosceles trapezium.



(Parallel sides denoted by Arrows)

Property - 1:

In an isosceles trapezium, the interior angles on the same side of each of the non-parallel sides are supplementary.

$$\therefore \angle A + \angle D = 180^\circ \quad (\text{as } AB \parallel CD \text{ and } AD \text{ is a transversal}).$$

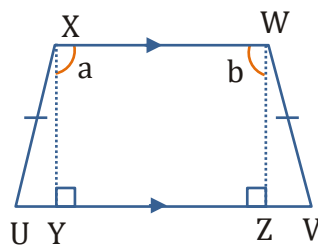
Property - 2:

In an isosceles trapezium, the angles opposite to the equal sides are equal.

To prove: $\angle U = \angle V$ and $\angle X = \angle W$

Construction: Draw XY and WZ perpendicular to UV

Proof:



Since $XW \parallel UV$,

$$a = 180^\circ - \angle XYZ = 90^\circ, \text{ and } b = 180^\circ - \angle WZY = 90^\circ$$

(since the internal angles on the same side of a transversal add up to 180°)

Hence, XWZY is a rectangle.

Now, In $\triangle UXY$ and $\triangle VWZ$

$$\angle XYU = \angle WZV \quad (\text{each } 90^\circ)$$

$$XY = WZ \quad (\text{distance between two parallel lines})$$

$XU = WV$ (by definition of isosceles trapezium)

$\triangle UXY \cong \triangle VWZ$ (by RHS congruency)

$\angle U = \angle V$ (by cpct congruency)

Also, $\angle UXY = \angle VWZ$... (i) (by cpct congruency)

Also, $\angle YXW = \angle ZWX$... (ii) ($a = b$ as each 90°)

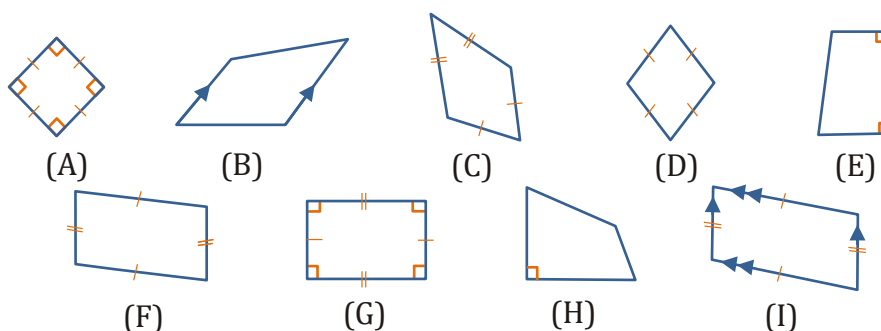
So, adding (i) and (ii)

$$\angle UXY + \angle YXW = \angle VWZ + \angle ZWX$$

$\angle UXW = \angle VWX$ or $\angle X = \angle W$ Hence proved.



Classify the quadrilateral using the name that best describes it.



Explanation

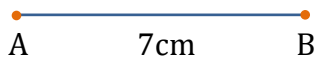
- (A) Since, all sides and all angles are equal. Hence, it is a square.
- (B) Since, a pair of opposite sides are parallel. Hence, it is a trapezium.
- (C) Since, one pair of adjacent sides are equal to each other and the other pair of adjacent sides are equal to each other. Hence, it is a kite.
- (D) Since, all sides are equal in this quadrilateral. Hence, it is a rhombus.
- (E) Since, adjacent angles are of 90° , opposite sides are parallel. Hence, it is a trapezium.
- (F) Since, in this quadrilateral both the pairs of opposite sides are equal in length. Hence, it is a parallelogram.
- (G) Since, opposite sides are equals and all angles are of 90° . Hence, it is a rectangle.
- (H) Since, it has 4 sides and 4 vertices and has no other special property. Hence, it is a quadrilateral.
- (I) Since, opposite sides are equal and parallel. Hence, it is a parallelogram.



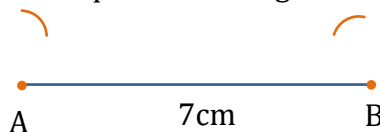
Construct a trapezium ABCD in which $AB = 7$ cm, $AD = 5$ cm, $BC = 6$ cm, and AB is parallel to DC . Write the steps of construction and draw the figure.

Explanation

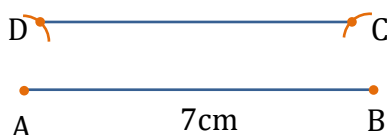
Step-1: Draw a base line $AB = 7$ cm



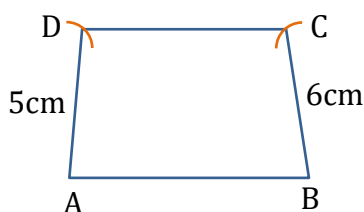
Step-2: Draw an arc from point A and point B of length 5 cm and 6 cm respectively.



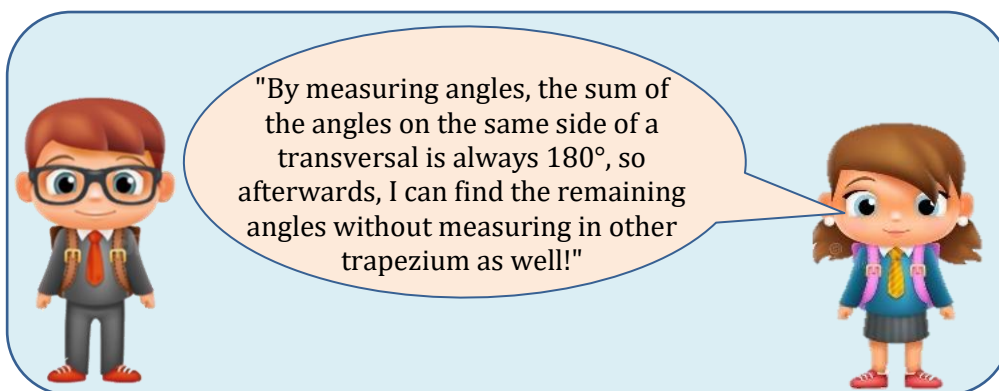
Step-3: Take a point D such that $AD = 5$ cm and from D, draw a line such that $DC \parallel AB$.



Step 4: Join AD and BC.



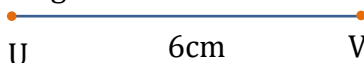
The quadrilateral ABCD formed is a trapezium.



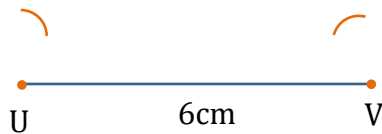
Construct an isosceles trapezium UVWX, with $UV \parallel WX$ where $UV = 6$ cm and $UX = VW = 5$ cm.

Explanation

Step 1: Draw UV of any suitable length.



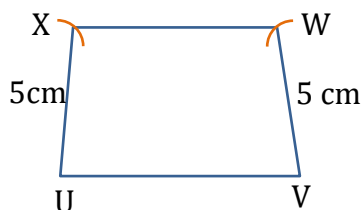
Step 2: Draw an arc from point U and V of length 5 cm



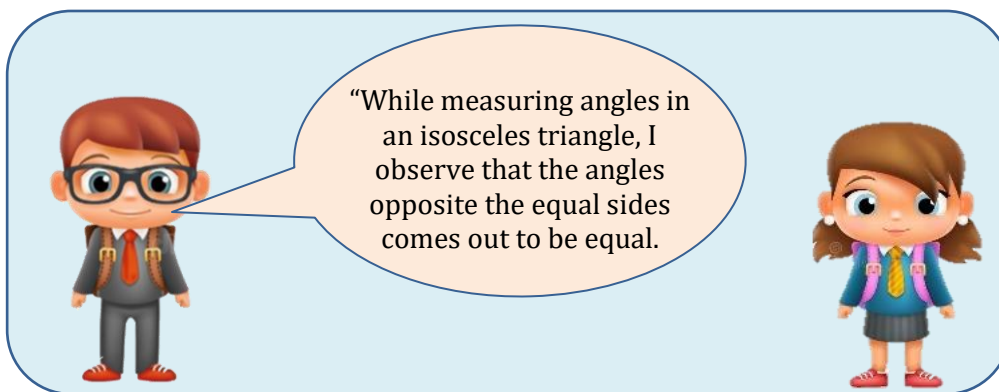
Step 3: Consider a point X on first arc and through it, draw $XW \parallel UV$



Step 4: Join UX and VW.

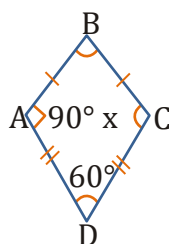


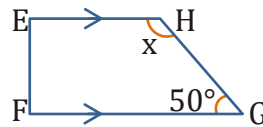
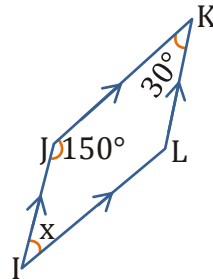
The quadrilateral formed is an isosceles trapezium.



Find the value of x in each quadrilateral.

(i) Kite ABCD



(ii) Trapezium EFGH**(iii) Parallelogram IJKL****Solution**

- (i) We know, in a kite, the opposite angles that are between the sides of unequal length are equal.

Here, $AB = BC$ and $AD = CD$

So, $\angle A = \angle C$ ($\angle A$ - between AB and AD , $\angle C$ - between BC and CD)

$$\Rightarrow \angle C = \angle A = x = 90^\circ$$

- (ii) We know, in a trapezium, the interior angles on the same side of each of the non-parallel sides are supplementary.

Here, $EH \parallel FG$

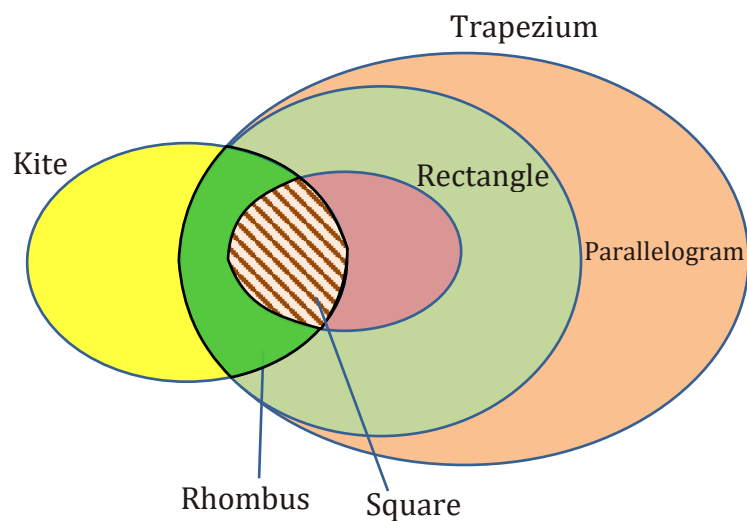
$$\Rightarrow x + 50^\circ = 180^\circ$$

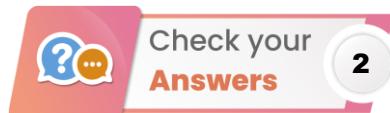
$$\Rightarrow x = 130^\circ$$

- (iii) We know, in a parallelogram, opposite angles are equal.

So, $\angle I = \angle K \Rightarrow x = 30^\circ$.

Hence, the complete Venn Diagram is as follows:



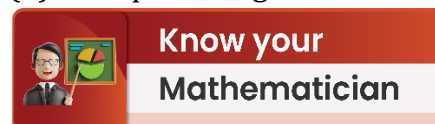


- (ii) 4 Right angles



Determine whether each statement is sometimes, always or never true.

- (i) A quadrilateral is a trapezium.
 (ii) A parallelogram is a rectangle.

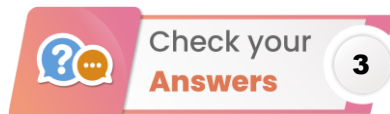


Euclid

Euclid, known as the *Father of Geometry*, laid the foundation of geometry in his book *Elements*. He studied shapes like triangles and quadrilaterals and defined their properties using logical reasoning. His work helps us understand angles, sides, and the classification of quadrilaterals.



Euclid



- (i) Sometime true (ii) Sometime true

Summary

- A **rectangle** has four right angles, opposite sides equal and parallel, and equal diagonals that bisect each other.
- A **square** has all sides equal, all angles 90° , and diagonals that are equal, bisect each other at 90° , and bisect the angles.
- A **parallelogram** has opposite sides equal and parallel, opposite angles equal, adjacent angles adding to 180° , and diagonals that bisect each other.
- A **rhombus** has all sides equal, opposite sides parallel, opposite angles equal, and diagonals that bisect each other at 90° and the angles.
- A **kite** has two pairs of equal adjacent sides and diagonals that intersect at 90° , with one diagonal bisected.
- A **trapezium** has at least one pair of opposite sides parallel.
- The **sum of all interior angles** in any quadrilateral is 360° .

Memory Map

Understanding Quadrilaterals

Curve

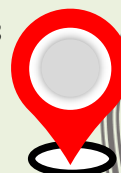


Curves can be of different types - open, closed, simple closed etc.

Classification of Polygons

A polygon is a simple closed curve made of many line segments.

Polygons



Quadrilateral



Pentagon



Hexagon



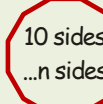
Heptagon



Octagon



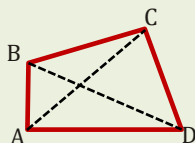
Nonagon



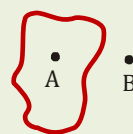
Decagon n-gon

Diagonals

AC and BD - diagonals



EH, HF, FI, IG, GE - diagonals



Region

Point A lies in the interior region of the curve and point B lies in the exterior region of the curve.

Regular and Irregular polygons

Name

Regular

Irregular

Triangle



Quadrilateral



Pentagon



Hexagon

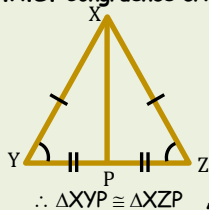


Rules of congruency

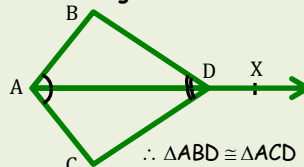
Rule-1: S.S.S. Congruence criterion



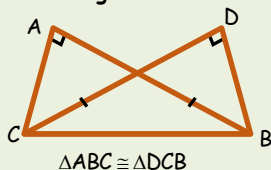
Rule-2: S.A.S. Congruence criterion



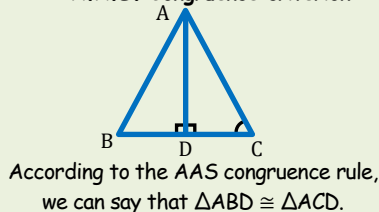
Rule-3: A.S.A. Congruence criterion



Rule-4: R.H.S. Congruence criterion



Rule-5: A.A.S. Congruence criterion



Sum of the measure of the exterior angles of a polygon

Quadrilateral ABCD.

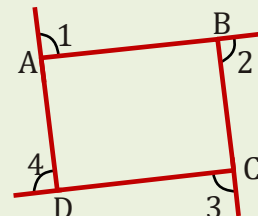
Sum of all the three angles of a triangle = 180°

The sum of the exterior angles of any polygon is equal to 360° .

Angle sum property of a quadrilateral

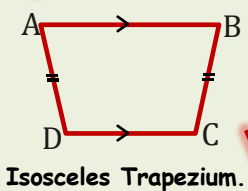
$$\Rightarrow \angle 1 + \angle 2 + \angle 3 + \angle 4 = 360^\circ$$

(a polygon with 4 sides) $\rightarrow$
Sum of all the four angles of a quadrilateral = 360°

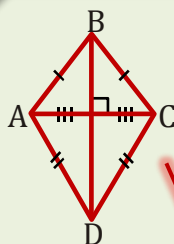


Kinds of Quadrilaterals

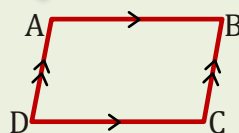
Trapezium



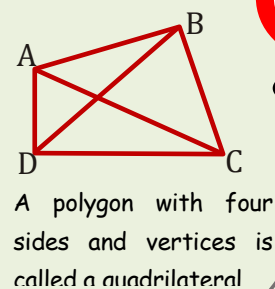
Kite



Parallelogram

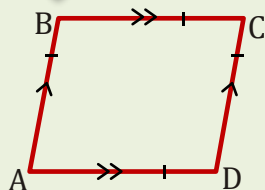


Quadrilaterals

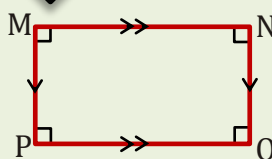


Some special parallelograms

Rhombus



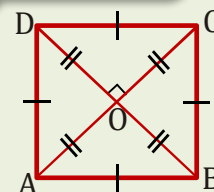
Rectangle



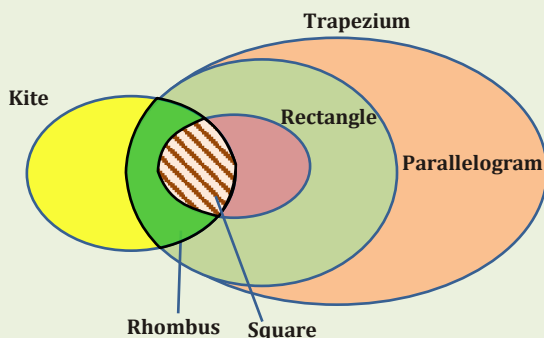
$$\angle M = \angle N = \angle O = \angle P = 90^\circ$$

The rectangle has all the properties of a parallelogram and a few more too.

Square



ABCD is a square. It has $AB = BC = CD = DA$ and $\angle A = \angle B = \angle C = \angle D = 90^\circ$



Venn diagram